

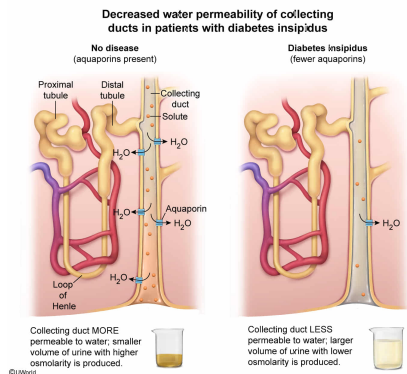
Patients with central diabetes insipidus (CDI) excrete large volumes of urine with low osmolarity. Given this, the nephrons of CDI patients most likely possess:

- ☐ A. loops of Henle that penetrate into the inner region of the medulla. [7%]
- ✓ ☒ B. collecting ducts composed of epithelial cells deficient in functional aquaporins. [69%]
- ☐ C. large numbers of Na^+/K^+ -ATPases in the cells of the proximal tubules. [11%]
- ☐ D. highly active urea transporters in the distal portions of the collecting ducts. [11%]

Correct

68% answered correctly

Explanation:



The **kidneys**, which consist of functional units called **nephrons**, are divided into an outer (renal) cortex and inner "salty" medulla. First, in the renal cortex, blood is **filtered** through glomerular capillaries into the Bowman capsule of each nephron. Water and solutes are either reabsorbed from or secreted into this **filtered fluid** (filtrate) as it progresses through the nephron's proximal convoluted tubule, loop of Henle, distal convoluted tubule, and collecting duct. Finally, filtrate is excreted from the body as urine.

A primary function of the **collecting duct** is **reabsorption of water**. This process is regulated by **antidiuretic hormone**, which increases the insertion of **aquaporins** into the **plasma membranes** of collecting duct epithelial cells. As a result, collecting ducts become **more permeable** to water, *increasing reabsorption* of water from the filtrate and **decreasing the volume** (ie, water content) of **excreted urine**. Decreasing the water volume of urine also *increases* its solute concentration (ie, **osmolarity**).

This question states that patients with **central diabetes insipidus** (CDI) produce **large volumes** of **urine** with **low osmolarity** (ie, dilute urine). Because collecting duct aquaporins *reabsorb* water into the body, their production of excess dilute urine is likely due to **collecting ducts** that are **deficient in functional aquaporins**.

The other options would *increase* water reabsorption, *decreasing* (not increasing) urine volume:

- **(Choice A)** **Countercurrent multiplication** in the loop of Henle contributes to the formation of a **concentration gradient** such that water reabsorption from filtrate is *maximized*. Therefore, nephrons with *long* loops of Henle penetrating into the inner region of the medulla (ie, **juxtamedullary nephrons**) would *increase* water reabsorption.
- **(Choice C)** Proximal tubule Na^+/K^+ -ATPases (ie, **sodium-potassium pumps**) **actively transport** sodium ions from the filtrate to the extracellular region *outside* the tubule. This allows *more* water to be reabsorbed as it flows via **osmosis** from the *less* concentrated filtrate in the tubule to areas of *higher* solute concentration outside the tubule.
- **(Choice D)** Urea transporters in the distal collecting duct reabsorb urea (solute) from the filtrate *into* the medulla. This *increases* medullary solute concentration and therefore *increases* the reabsorption (**osmosis**) of water *from* the filtrate *into* the salty medulla.

Educational objective:

When water permeability of the kidneys' collecting ducts is increased via insertion of aquaporins into collecting duct cell membranes, urine output decreases because more water is reabsorbed into the body.

Time spent: 0 sec
Subject:
Biology

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System:
Digestion and Excretion